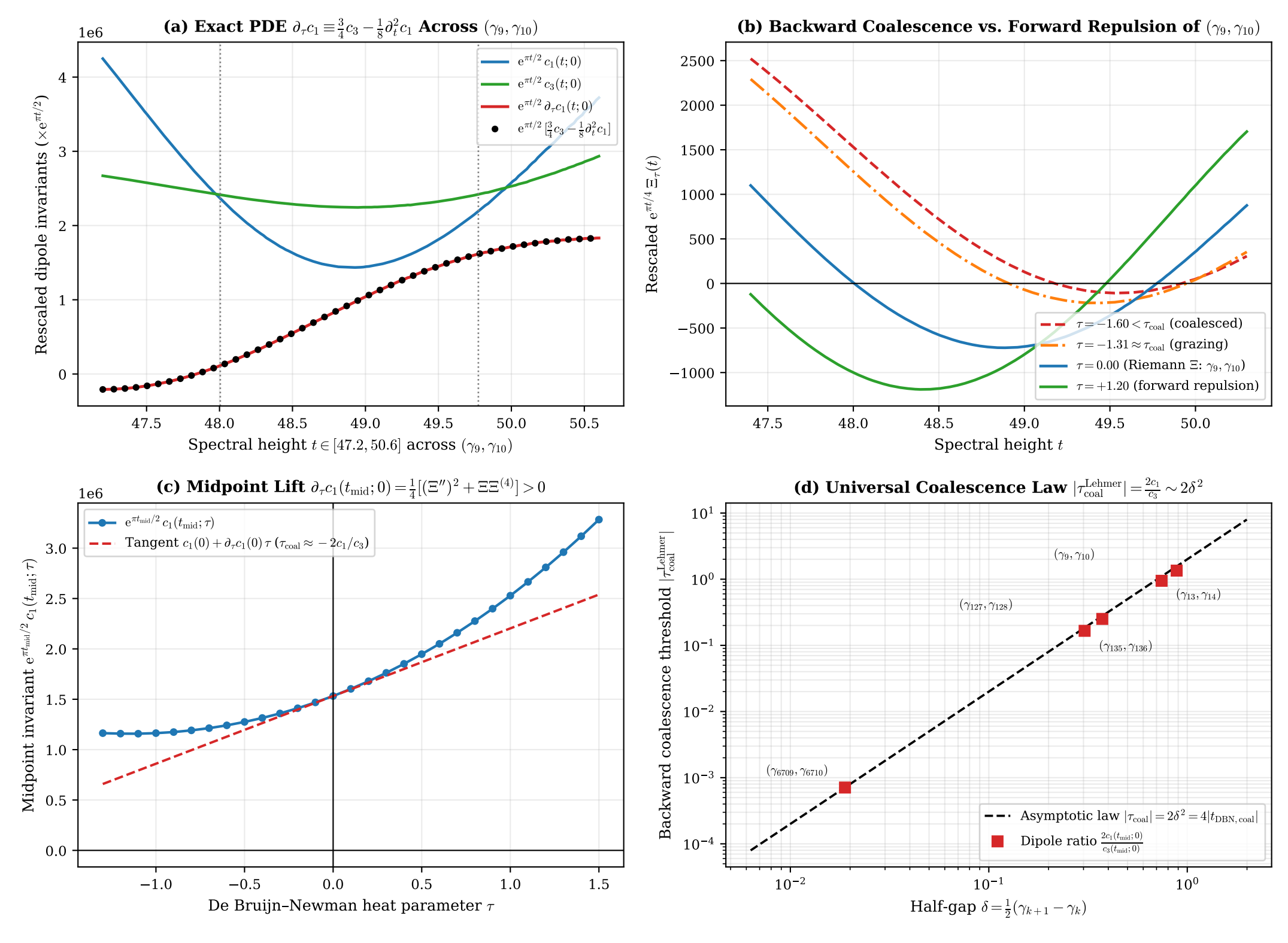

(a) Exact PDE $\partial_\tau c_1 \equiv \frac{3}{4}c_3 - \frac{1}{8}\partial_t^2 c_1$ Across (γ_9, γ_{10})
1e6
Rescaled dipole invariants ($\times e^{\pi t/2}$)
$e^{\pi t/2}\, c_1(t;0)$
$e^{\pi t/2}\, c_3(t;0)$
$e^{\pi t/2}\, \partial_\tau c_1(t;0)$
$e^{\pi t/2}\, [\frac{3}{4}c_3 - \frac{1}{8}\partial_t^2 c_1]$
Spectral height $t \in [47.2, 50.6]$ across (γ_9, γ_{10})
(b) Backward Coalescence vs. Forward Repulsion of (γ_9, γ_{10})
Rescaled $e^{\pi t/4}\, \Xi_\tau(t)$
$\tau = -1.60 < \tau_{\text{coal}}$ (coalesced)
$\tau = -1.31 \approx \tau_{\text{coal}}$ (grazing)
$\tau = 0.00$ (Riemann Ξ: γ_9, γ_{10})
$\tau = +1.20$ (forward repulsion)
Spectral height t
(c) Midpoint Lift $\partial_\tau c_1(t_{\text{mid}};0) = \frac{1}{4}[(\Xi'')^2 + \Xi\Xi^{(4)}] > 0$
1e6
Midpoint invariant $e^{\pi t_{\text{mid}}/2}\, c_1(t_{\text{mid}};\tau)$
$e^{\pi t_{\text{mid}}/2}\, c_1(t_{\text{mid}};\tau)$
Tangent $c_1(0) + \partial_\tau c_1(0)\,\tau$ ($\tau_{\text{coal}} \approx -2c_1/c_3$)
De Bruijn–Newman heat parameter τ
(d) Universal Coalescence Law $|\tau_{\text{coal}}^{\text{Lehmer}}| = \frac{2c_1}{c_3} \sim 2\delta^2$
Backward coalescence threshold $|\tau_{\text{coal}}^{\text{Lehmer}}|$
(γ_9, γ_{10})
$(\gamma_{13}, \gamma_{14})$
$(\gamma_{127}, \gamma_{128})$
$(\gamma_{135}, \gamma_{136})$
$(\gamma_{6709}, \gamma_{6710})$
Asymptotic law $|\tau_{\text{coal}}| = 2\delta^2 = 4|t_{\text{DBN,coal}}|$
Dipole ratio $\frac{2c_1(t_{\text{mid}};0)}{c_3(t_{\text{mid}};0)}$
Half-gap $\delta = \frac{1}{2}(\gamma_{k+1} - \gamma_k)$